



SALAS O'BRIEN

3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Riverbrook 9; Beltan Properties
Location: Gainesville, GA
Description: Tranquility - UBSmt, SW-Facing

Date: March 17, 2024
By: JTH / TJH
House Faces: SouthWest

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	2,741 ft ²
Volume	25,619 ft ³
Percentage Glass	14 %
Number of Bedrooms	4

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	12621
SEER	14
ACH50	5.0
Mechanical Ventilation	NO
House Outside Air CFM	N/A

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.048	Basement	Concrete + 2x6	R-19 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.167	Exterior Door		R-6	
	D2	0.5	Garage Door		R-2	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling		R-38 sprayed	
	R2	0.053	Vaulted		R-19 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.077	Framed		R-13 sprayed	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.30	Blinds/Draperies	SHGC 0.18
	G2					
	G3					

LOAD SUMMARY					
			Unit 1 - Whole House		
			Cooling	Heating	
Sensible Load			31227 Btu/hr	45439 Btu/hr	
Minimum Nominal Capacity			3.47 Tons	13.31 kW	
Load Density			790 SF/Ton	4.9 W/SF	
Floor Area Served			2741.0 SF	2741.0 SF	
Mechanical Ventilation			N/A CFM	N/A CFM	
System Size			3.5 Tons	15.0 kW	
Leaving Air Temperature			59.8 deg F	99.6 deg F	
Installed Capacity			783 SF/Ton	5.5 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

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[illegible]

Unit 1

Size

3.50

Tons

Airflow

1400

CFM

Heat kW

15

kW

First Floor Airflows

Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Foyer / Hall	54	74	64	
Master WIC	82	94	88	
Master Bath	82	92	87	
Master Suite	121	145	133	
Family	235	210	223	
Kitchen	151	47	99	
Dining	110	165	137	
Laundry	57	32	44	
Garage Entry Hall	44	59	52	
Bath 2	26	26	26	
Bed 2 / Study	91	100	95	
-	-	-	-	
-	-	-	-	
-	-	-	-	

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Cooling Size	3	DX Cooling kWh	3130
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By: JTH / TJH

Heating Size	3	RC Heating kWh	6303
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Best Case	2.5	Total kWh	9432
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SEER 14

COOLING LOAD CALCULATION										Ceiling Height		9		9		9		9		9		9		9		9		9		9		9			
Components					CLTD (°F)	U-Value (Btu/hr-sq-F)	Cooling Load Factor	Entire House		Foyer / Hall		Master WIC		Master Bath		Master Suite		Family		Kitchen		Dining		Laundry		Garage Entry Hall		Bath 2		Bed 2 / Study					
	Type	Description		Qty				Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty
Glass	G1	SouthEast	Right	(E)	12	Btu/hr-sf	18	213													9	107	4.5	53	4.5	53									
	G1	South		(SE)	13	Btu/hr-sf	0	0																											
	G1	SouthWest	Front	(S)	22	Btu/hr-sf	25	539			25	539																							
	G1	West		(SW)	24	Btu/hr-sf	0	0																											
	G1	NorthWest	Left	(W)	15	Btu/hr-sf	56	854					37.5	570	18.64	283																			
	G1	North		(NW)	5	Btu/hr-sf	0	0																											
	G1	NorthEast	Back	(N)	7	Btu/hr-sf	96	630							36	236					60	394													
	G1	East		(NE)	11	Btu/hr-sf	0	0																											
	G1	Sky Light			38	Btu/hr-sf	0	0																											
Door	G1	Shaded			5	Btu/hr-sf	148	776	17.81	93						66.02	346																		
	D1	Door		21	0.17	3.51	Btu/hr-sf	28	98	17.81	62					10.01	35									17.81	267								
	D2	Door		30	0.50	15.00	Btu/hr-sf	18	267																										
	W1	SouthEast		21	0.08	1.62	Btu/hr-sf	394	637			65	105								115	186	93	150	121	195									
	W1	South		16	0.08	1.23	Btu/hr-sf	0	0																										
	W1	SouthWest		21	0.08	1.62	Btu/hr-sf	272	440	37	60	141	228															95	153						
	W1	West		23	0.08	1.77	Btu/hr-sf	0	0																										
	W1	NorthWest		19	0.08	1.46	Btu/hr-sf	461	674			70	102	128	187	166	242				48	70							50	73					
	W1	North		13	0.08	1.00	Btu/hr-sf	0	0																										
Basement Walls	W1	NorthEast		19	0.08	1.46	Btu/hr-sf	498	729						108	158	262	384			128	187													
	W1	East		23	0.08	1.77	Btu/hr-sf	0	0																										
	W1	SouthEast		21	0.08	1.62	Btu/hr-sf	0	0																										

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Level(s): Second Floor
SEER 14

Cooling Size	1	DX Cooling kWh	1034
Heating Size	1	RC Heating kWh	2155
Best Case	1	Total kWh	3189

COOLING LOAD CALCULATION										Ceiling Height		19		9		9		9		9		9		9		9		9		9		9	
Components					CLTD (°F)	U-Value (Btu/hr-sq-F)	Cooling Load Factor	Entire House		Stairs		Hallway		Bed 3 WIC		Bed 3		Bath 4		Bed 4		Bed 4 WIC											
	Type	Description	Qty	Btu/hr				Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr
Glass	G1	SouthEast	Right	(E)	12	Btu/hr-sf	30	355		-		-		-	30	355		-		-		-		-		-		-		-		-	
	G1	South		(SE)	13	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	SouthWest	Front	(S)	22	Btu/hr-sf	15	324		-		-		-		-	15	324		-		-		-		-		-		-		-	
	G1	West		(SW)	24	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	NorthWest	Left	(W)	15	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	North		(NW)	5	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	NorthEast	Back	(N)	7	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	East		(NE)	11	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	Sky Light			38	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-	
G1	Shaded			5	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		
Door	D1	Door		21	0.17	3.51	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-
	D2	Door		15	0.50	7.50	Btu/hr-sf	20	150		-		-		-		-		-		-		-		-		-		-		-		-
Above Grade Walls	W1	SouthEast		21	0.08	1.62	Btu/hr-sf	128	206	-	-	-	-	10	75	112	181	16	26	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	South		16	0.08	1.23	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	W1	SouthWest		21	0.08	1.62	Btu/hr-sf	122	197	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	W1	West		23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	W1	NorthWest		19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	W1	North		13	0.08	1.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	W1	NorthEast		19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	W1	East		23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	W1	SouthEast		21	0.08	1.62	Btu/hr-sf	0	0</																								